

Sushant S. Mahajan

PROFESSIONAL PREPARATION

Postdoctoral Fellow, Stanford University	2022–present
Solar Physics Postdoctoral Fellow, University of Hawaii	2019–2022
<i>Ph.D.</i> in Astronomy, Georgia State University	2019
<i>M.S.</i> in Physics, Georgia State University	2014–2017
<i>M.Tech.</i> in Engineering Physics, IIT (BHU)	2014

RESEARCH PROFILE

Solar physicist and computational researcher with expertise in solar dynamo diagnostics, global-scale solar flows, time-series analysis, rare-event prediction, and scientific software for large observational datasets.

HONORS

Stanford University Libraries Data Sharing Prize, contributing member of the Stanford Solar Observatories Group	2026
Editor’s Choice, <i>Solar Physics</i>	2024
Best Young Presenter Award, IAU Symposium 340	2018
Best Young Scientist Poster Award, IAU Symposium 335	2017
Honorary Mention, SPD student poster competition	2016

SELECTED PUBLICATIONS

Mahajan, S. S. , et al. <i>The Sun’s Large-Scale Flows I: Measurements of Differential Rotation & Torsional Oscillation</i> . <i>Solar Physics</i>	2024
Mahajan, S. S. , Sun, X., and Zhao, J. <i>Removal of Active Region Inflows Reveals a Weak Solar Cycle Scale Trend in the Near-Surface Meridional Flow</i> . <i>ApJ</i>	2023
Pal, S., Bhowmik, P., Mahajan, S. S. , and Nandy, D. <i>Impact of Anomalous Active Regions on the Large-Scale Magnetic Field of the Sun</i> . <i>ApJ</i>	2023
Hathaway, D. H., Upton, L. A., and Mahajan, S. S. <i>Variations in Differential Rotation and Meridional Flow within the Sun’s Surface Shear Layer 1996–2022</i> . <i>Frontiers in Astronomy and Space Sciences</i>	2022
Mahajan, S. S. , Hathaway, D. H., Munoz-Jaramillo, A., and Martens, P. C. <i>Improved Measurements of the Sun’s Meridional Flow and Torsional Oscillation from Correlation Tracking on MDI & HMI Magnetograms</i> . <i>ApJ</i>	2021
Ahmadzadeh, A., et al. <i>How to Train Your Flare Prediction Model: Revisiting Robust Sampling of Rare Events</i> . <i>ApJS</i>	2021
Angryk, R. A., et al. <i>Multivariate Time Series Dataset for Space Weather Data Analytics</i> . <i>Scientific Data</i>	2020

SELECTED TALKS

KIPAC public science seminar, Stanford University	2023
AGU Fall Meeting oral presentation on active-region inflows and meridional flow	2022

TESS oral presentation on solar-cycle trends in meridional flow	2022
AGU oral presentation on parameterization of inflows around active regions	2022
SPD/AAS oral presentation on active-region inflows and cycle 24 meridional flow	2021
AGU oral presentation on empirical relationships in sunspot number and area time series	2020

TEACHING, MENTORING, AND OUTREACH

Graduate Teaching Assistant, Georgia State University	2014–2018
Advised Ansh Malviya on statistical analysis of solar flares	2023
Mentored Prabhanjan Kulkarni on solar flare forecasting models	2020–2021
Lecturer, HI STAR outreach program, University of Hawaii	2020–2021
Volunteer at Stanford eclipse events and prior public observatory outreach programs	2014–present

COMPUTATIONAL TOOLS

Python, MATLAB, Fortran, bash, OpenMP, MPI, C++, CUDA, ParaView, scientific time-series analysis, and ML for forecasting applications.